Model	Method	Original Debate	Conditional Debate	Skip Rate
Llama-3.3 70B	MAD	79.04	**79.91**	59.83%
	Debate	80.35	**83.41**	68.56%
Qwen-2.5 72B	MAD	73.80	**76.86**	51.53%
	Debate	**79.91**	79.48	45.85%

Table 4: Results of applying conditional debate to existing debate systems on the StrategyQA dataset

this trade-off, the voting-based approach is preferable when prioritizing computational efficiency, as it achieves comparable accuracy with fewer computational resources.

5.2 Response Shifts in Accuracy

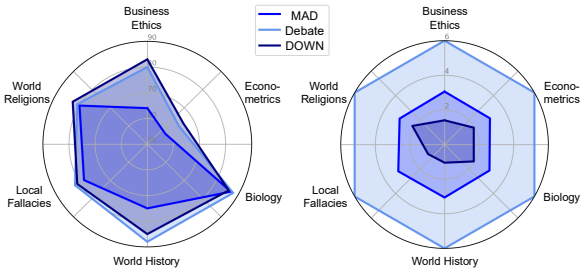
Table 3 reports the proportions of correct and incorrect changes in answers, computed over StrategyQA samples where the final answer differs from the initial prediction. To deepen our investigation, we set the confidence threshold to 0.9, increasing the number of cases where debate is triggered. Across all model configurations, DOWN consistently achieves a higher successful correction rate than baseline debate systems. Notably, for the LLaMA-3.3-70B model, 87.43% of changed answers reflect successful corrections of initially incorrect predictions. In contrast, MAD shows a high rate of incorrect changes, frequently revising correct initial answers into incorrect ones, consistent with the trends observed in § 5.1. The Debate baseline shows a similar rate of correct and incorrect changes, indicating a limited capacity to prioritize reliable peer inputs. In contrast, the high success rate of corrections achieved by DOWN stems from two core design principles. It initiates debate only when the model's confidence is low, thereby avoiding unnecessary changes to already correct answers. When debate is triggered, it refers to peer responses with their associated confidence scores, enabling the model to incorporate more reliable inputs.

5.3 Effects of Conditional Debate in Multiagent Debates

To assess the effectiveness of conditional debate, we apply it to MAD and Debate frameworks and evaluate performance on the StrategyQA dataset using models with approximately 70B parameters. Our findings in Table 4 reveal that applying conditional debate to existing debate methods mostly improves performance. In particular, the Debate framework achieves about a 3% point accuracy increase on LLaMA-3.3 70B, while MAD exhibits

Method	Acc.	AC
DOWN (θ^*)	71.18	2.53
Debate		
(1) w/o confidence-based scoring in debate	68.12	2.46
(2) w/o multi-agent debate (single response only)	70.74	1.00
(3) w/o debate skipping (debate only)	70.74	6.00
Threshold		
(4) w/ lowered threshold ($\theta^* - 0.1$)	69.87	1.20
(5) w/ raised threshold ($\theta^* + 0.1$)	71.62	3.53

Table 5: Ablation study on confidence score using the StrategyQA dataset



(1) Accuracy (2) Average Agent Calls

Figure 4: Accuracy and average agent calls (AC) of multiagent debate methods across six MMLU domains

a similar improvement on Qwen-2.5 72B. Despite these gains, debate skip rates remain high at 59.83% and 51.53%, respectively, demonstrating remarkable efficiency improvements. These results demonstrate that conditional debate not only enhances efficiency but also positively impacts overall model performance by selectively engaging discussions at appropriate points.

5.4 Ablation Study

To analyze the contribution of each component in the DOWN framework, we conduct an ablation study on the StrategyQA using the LLaMA-3.1 8B model. The results are shown in Table 5. We examine the role of confidence scores shared among agents during debate. Removing these signals leads to a 3.06% point drop in accuracy, indicating that confidence serves as an informative cue for selectively incorporating peer responses during the debate process. When the debate process is either entirely removed or enforced for every input, the accuracy drops to 70.74 in both cases. These findings indicate that always engaging in debate is redundant, yielding no improvement over the single agent baseline. The next analysis focuses on the threshold selection strategy. To assess its effectiveness, we vary the threshold by ± 0.1 and observe the resulting impact on performance. Modifying

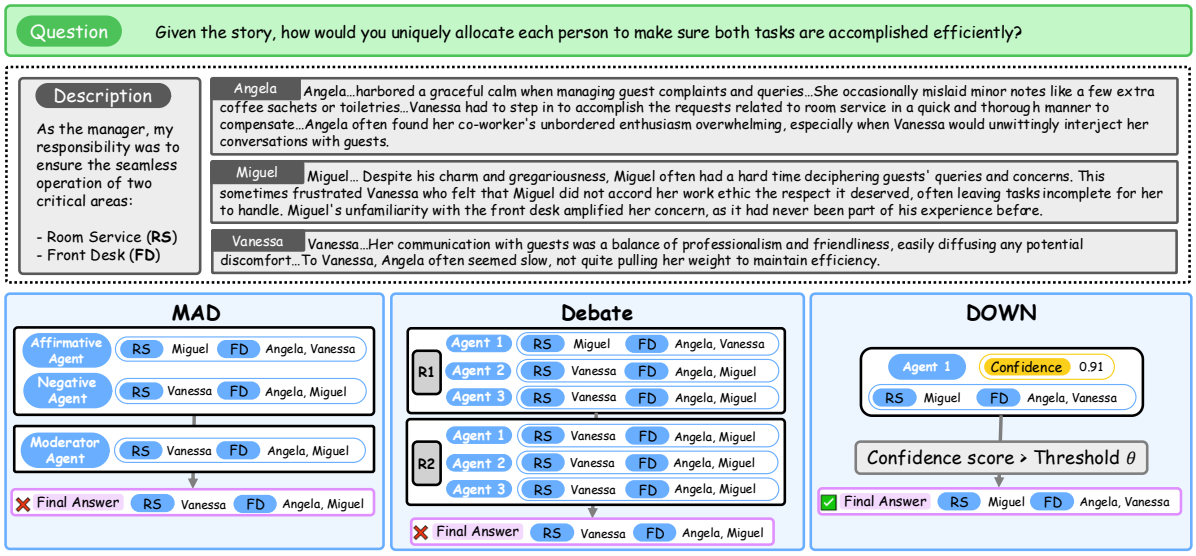


Figure 5: Qualitative analysis of the MUSR dataset

the threshold above or below the selected value reveals a trade-off between accuracy and average agent calls. Specifically, the scores obtained for the threshold values $\{\theta^* - 0.1, \theta^*, \theta^* + 0.1\}$ yields $\{0.29, 1.19, 1.00\}$, respectively. The chosen threshold θ^* yields the highest overall score, effectively balancing accuracy and efficiency. It maintains accuracy within a statistically reliable range, as determined by the Wilson lower bound, while substantially reducing computational cost.

5.5 Generalization analysis on MMLU

To assess the generalization ability of the DOWN framework across diverse task types and knowledge domains, we additionally evaluate its performance on MMLU using GPT-4o-mini. The performance of different multiagent debate systems across six domains is presented in Figure 4. Experimental results show that DOWN achieves performance comparable to that of the Debate method across tasks. Given that the Debate approach consistently requires an average of six agent calls, the DOWN framework shows significantly higher efficiency. The results suggest that DOWN generalizes well, achieving robust performance on benchmarks evaluating both reasoning capabilities and factual knowledge across diverse domains. This highlights its potential to serve as a scalable and effective alternative to full multiagent debate systems.

5.6 Qualitative Analysis

Figure 5 presents a qualitative comparison of responses generated by different debate systems on the MUSR dataset. The results show that existing debate mechanisms introduce unnecessary modifications. Although the initial responses produced by the Debate and MAD methods are correct, subsequent iterative revisions lead to an incorrect final prediction. This implies the potential risk of error propagation when redundant debate occurs. In contrast, with an initial confidence score of 0.91, DOWN skips the debate and directly adopts the initial response as the final answer. This suggests that selectively engaging in debate under high-confidence conditions prevents unnecessary modifications while maintaining efficiency. To further substantiate these findings, we present additional qualitative analysis in Appendix B.

6 Conclusion

This work proposed the DOWN framework to address the computational inefficiencies and error propagation challenges in multiagent collaboration. By conditionally activating debate based on the model confidence score, the approach significantly reduced computational overhead while preserving or even improving performance. The results showed that conditional debate enhanced efficiency and mitigated cascading errors, leading to more stable reasoning behavior. Furthermore, the confidence-guided multiagent debate amplified the influence of reliable responses on final decisions. These findings established DOWN as an effective optimization strategy, offering a high-performance and efficient solution for multiagent collaboration systems. multiagent collaboration systems.

Limitations

While our proposed DOWN framework demonstrates strong efficiency and robustness, several limitations remain. This study primarily focuses on English debates, which may limit its applicability to multilingual multiagent collaboration settings. Extending our method to multilingual LLMs would provide deeper insights into its robustness across diverse linguistic contexts. Due to computational constraints, we employ the GPT-4o-mini instead of the GPT-4o model. Applying the GPT-4o model to our system could further deepen the understanding of our DOWN framework.

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